

Rectangular Waveguide (TE₁₀)



$$\underline{E_x = 0}$$

$$\underline{H_y = 0}$$

$$\underline{E_z = 0}$$

$$\left[\begin{array}{l} E_y = -j C_1 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta' z} \\ H_x = +j C_2 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta' z} \\ H_z = C_3 \cos\left(\frac{\pi x}{a}\right) e^{-j\beta' z} \end{array} \right]$$

$$\left[\begin{array}{l} \text{Re}[E_y] = + C_1 \sin\left(\frac{\pi x}{a}\right) \sin(\beta' z) \\ \text{Re}[H_x] = + C_2 \sin\left(\frac{\pi x}{a}\right) \sin(\beta' z) \end{array} \right] \left[\begin{array}{l} \text{Re}[H_z] = C_3 \cos\left(\frac{\pi x}{a}\right) \cos(\beta' z) \end{array} \right]$$

Rectangular Waveguide (TE10)



$$H_n = \sin\left(\frac{\pi x}{a}\right) \sin(\beta' z)$$

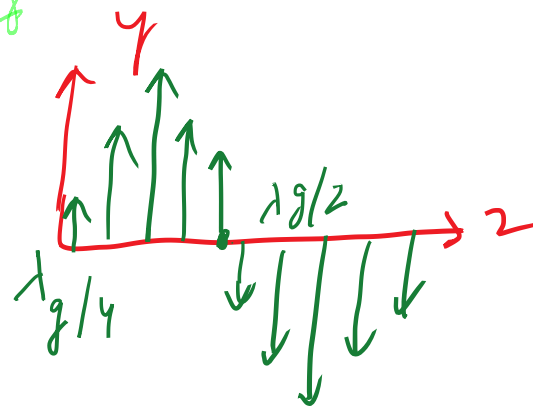
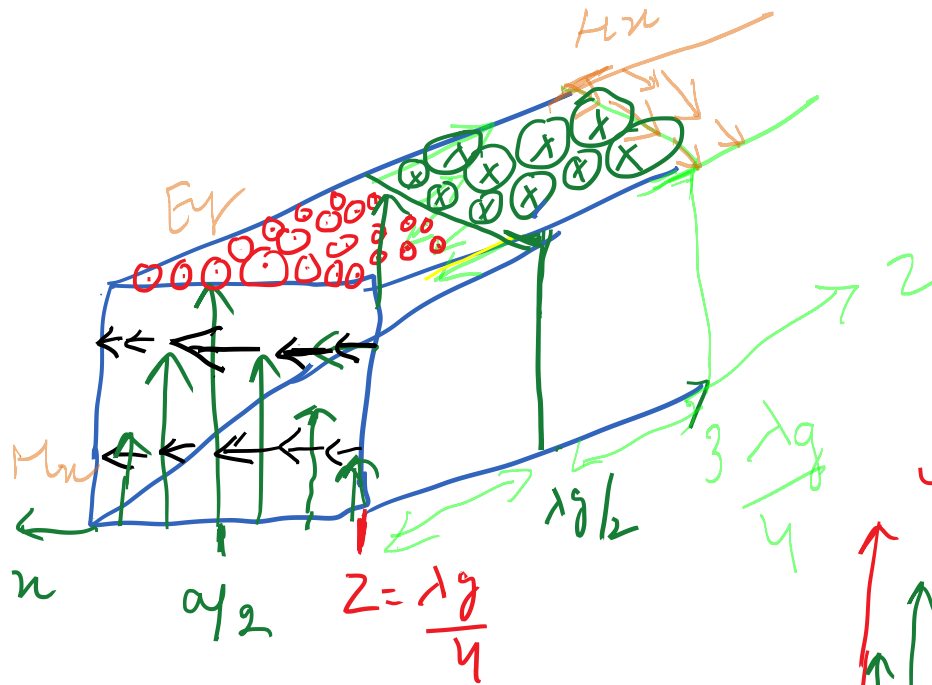
$$E_y = \sin\left(\frac{\pi x}{a}\right) \sin(\beta' z)$$

$$m = 1$$

$$\beta' z = \pi/2$$

$$z = \frac{\pi x a}{2 \times 2\pi}$$

$$z = \frac{a}{4}$$



at $a/4 \rightarrow H_n$ exists E_y exists $H_z = 0$

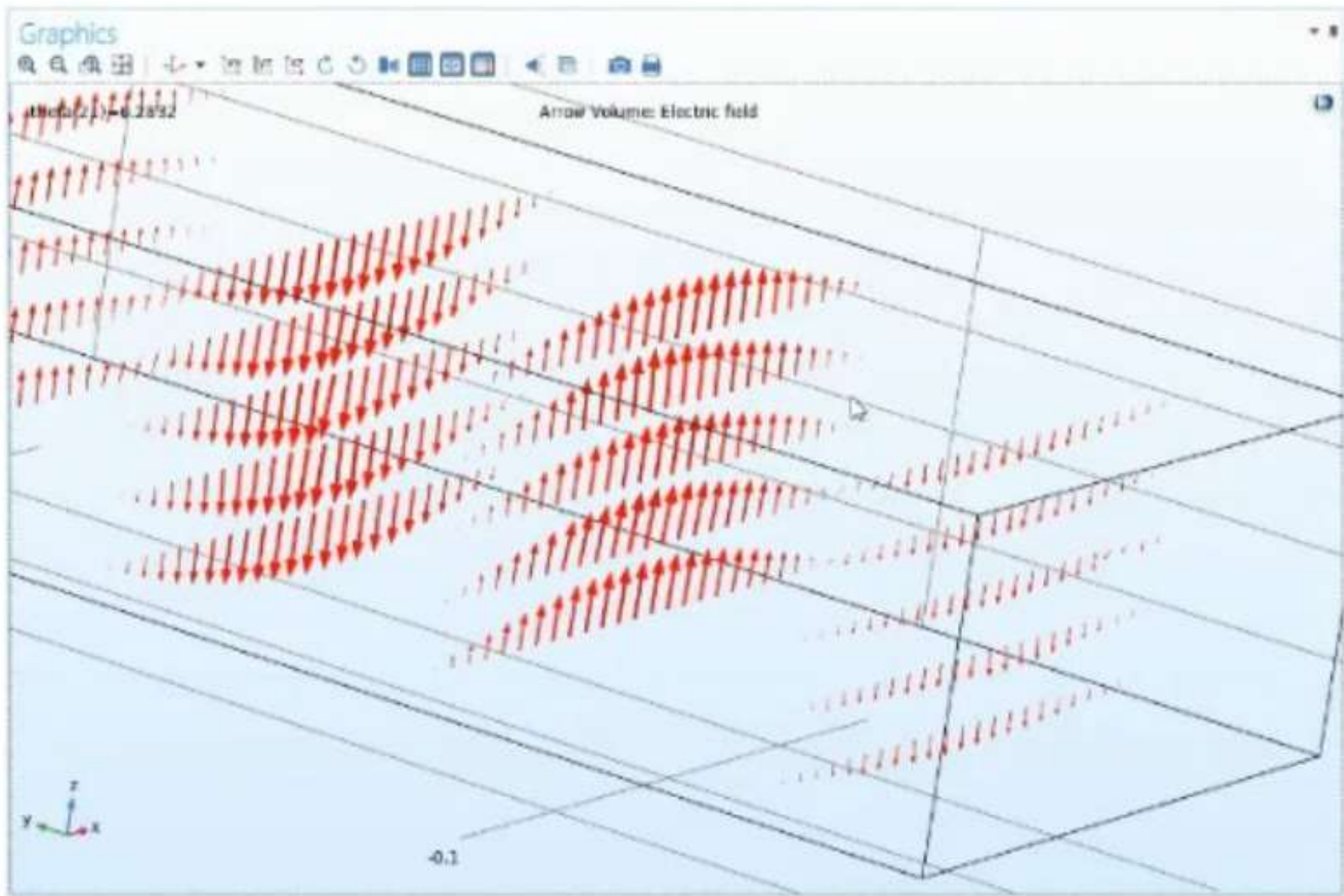
at $a/2 \rightarrow H_z$ exists E_y & $H_n = 0$

at $3a/4 \rightarrow H_n$ exists E_y exists $H_z = 0$

$$H_z = \cos\left(\frac{\pi x}{a}\right) \cos(\beta' z)$$

Ey field in rectangular waveguide

TE₁₀

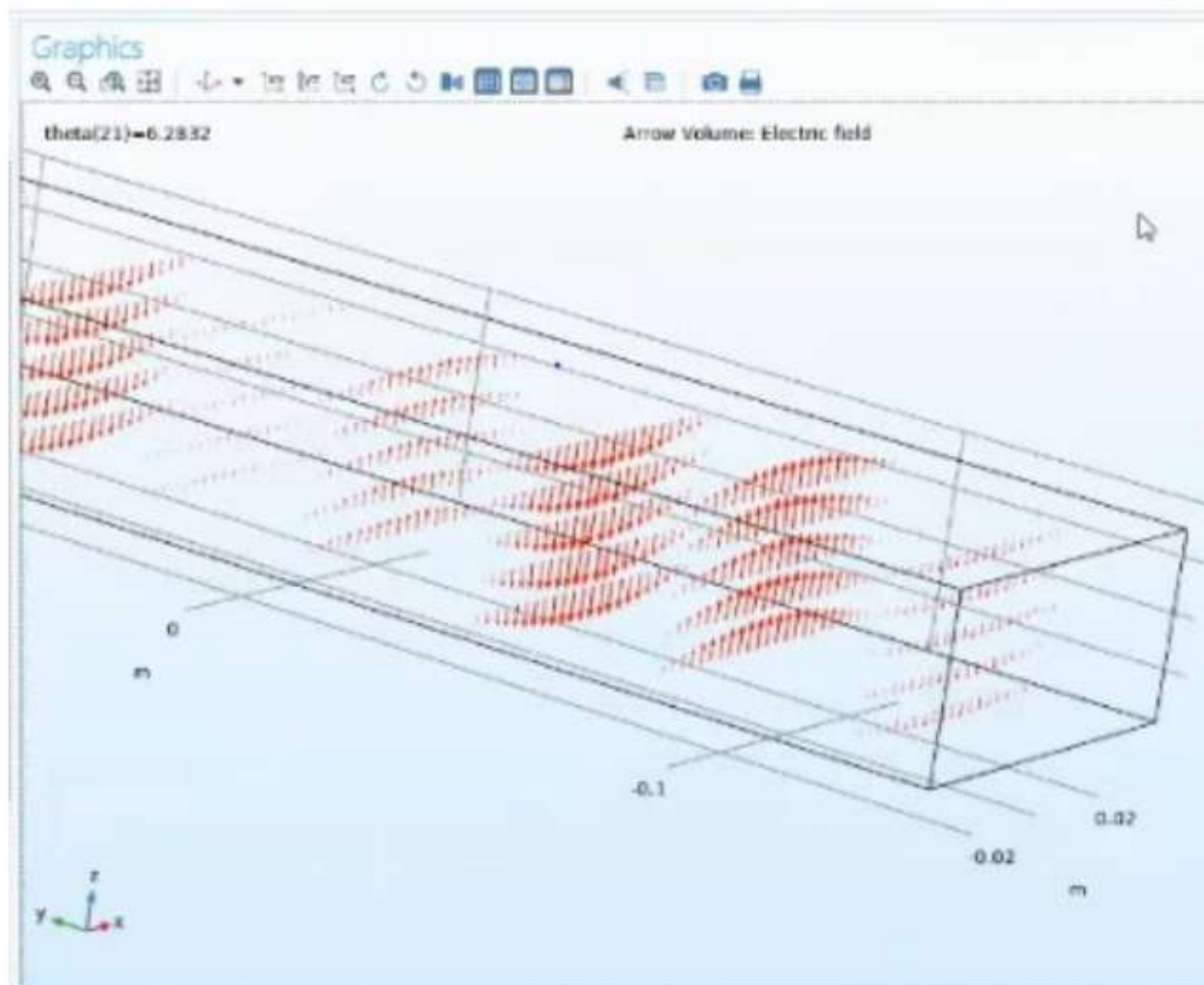


Ey field in rectangular waveguide TE₁₀

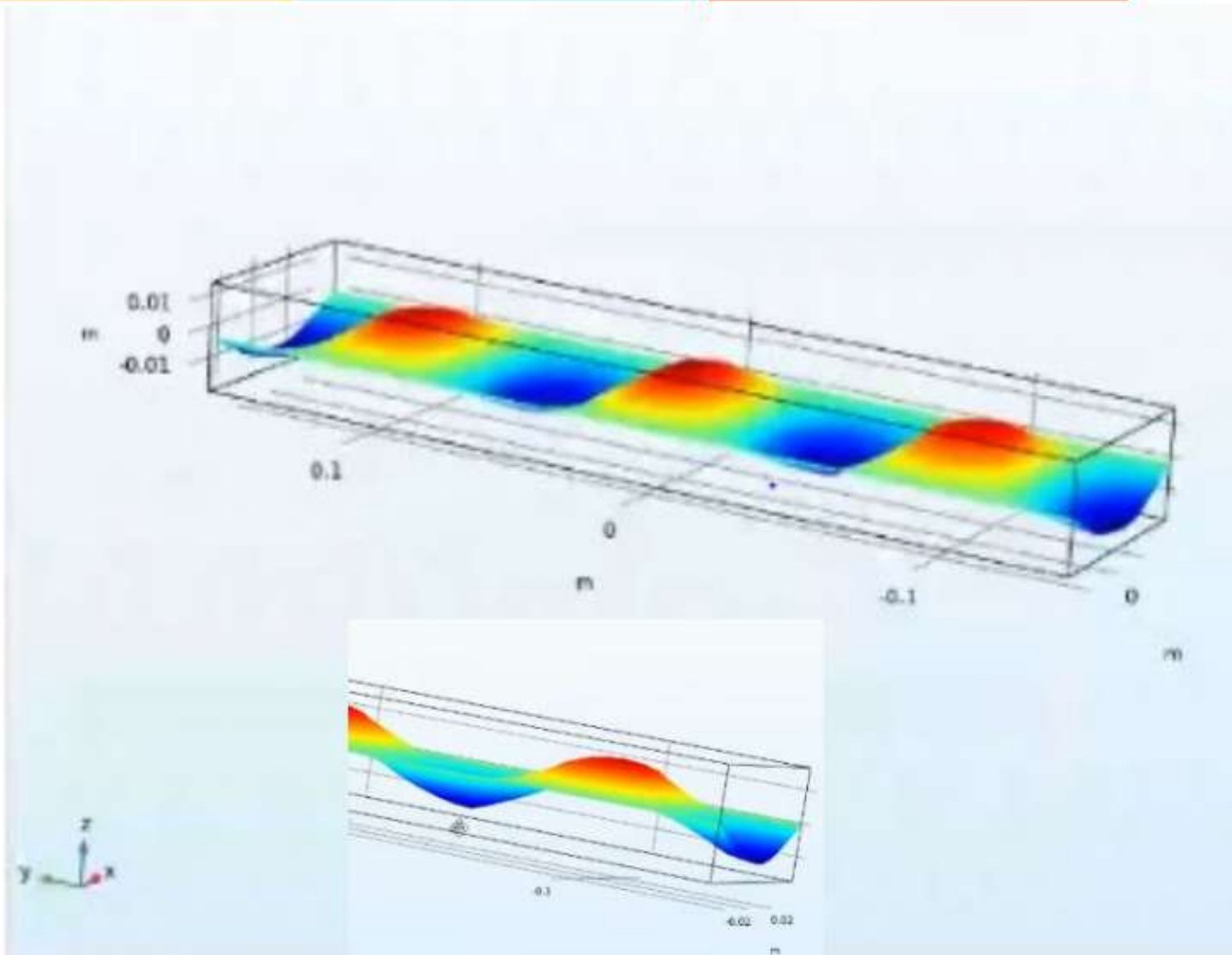
innovate

achieve

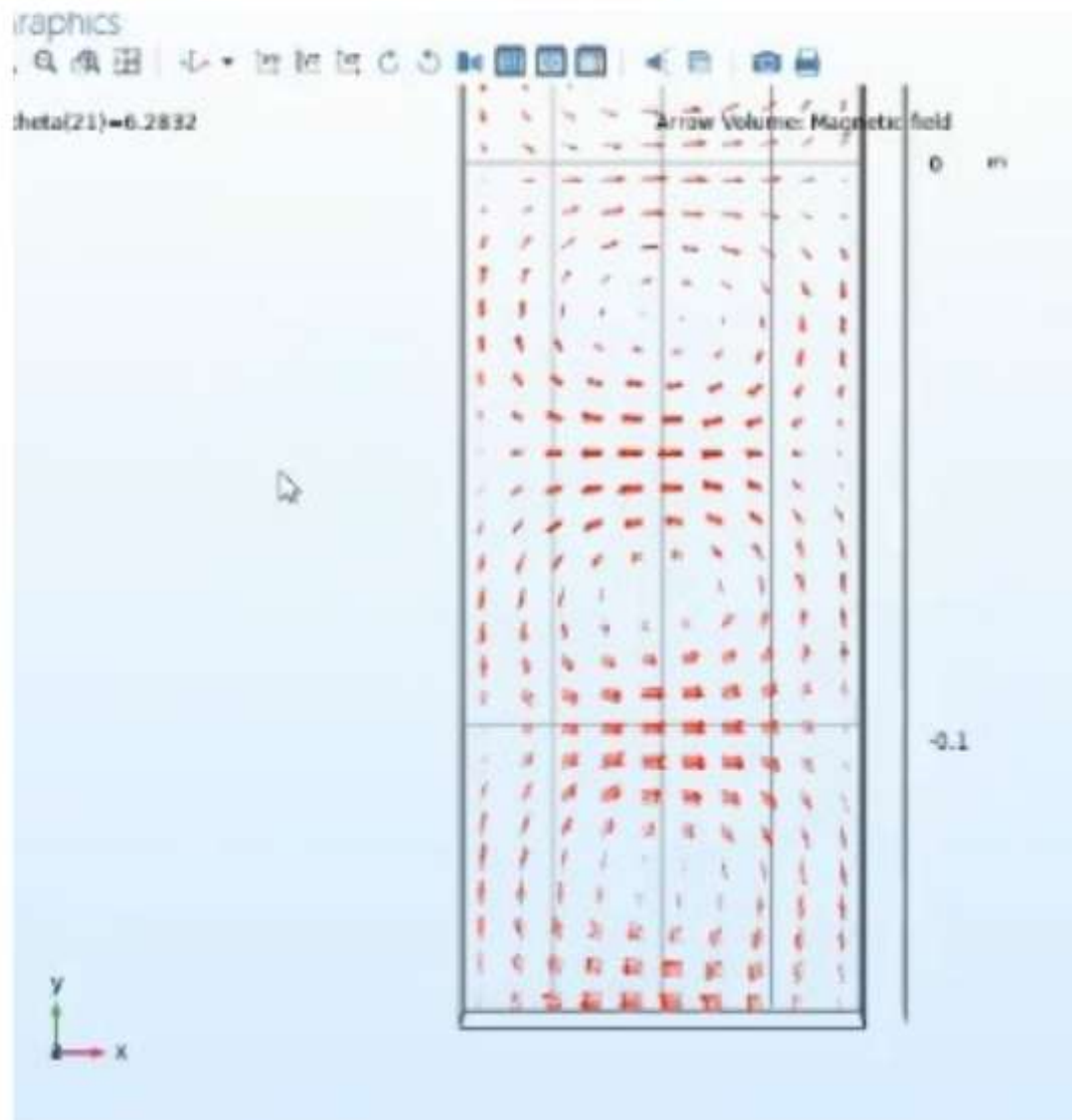
lead



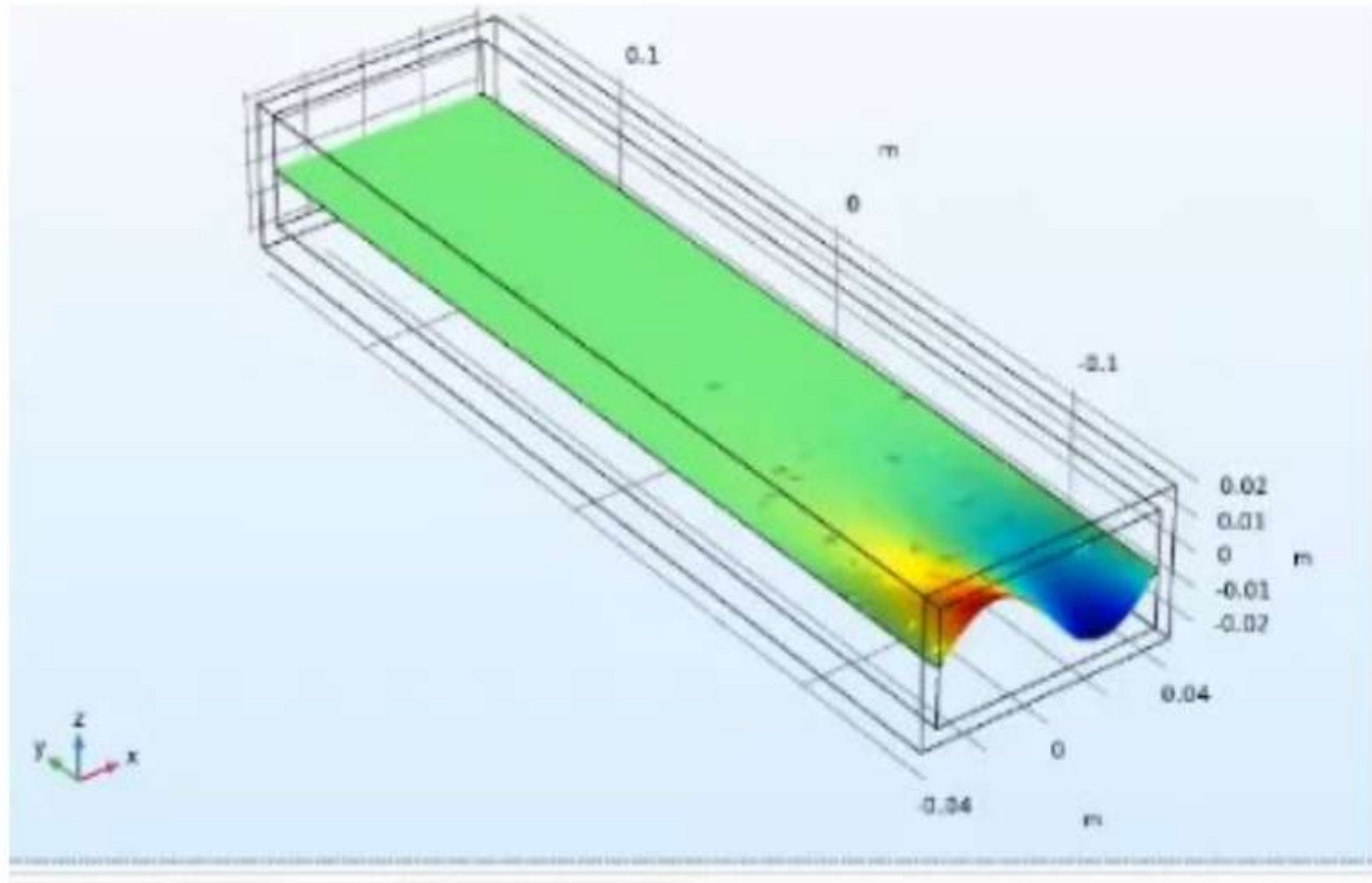
Ey field in rectangular waveguide TE₁₀



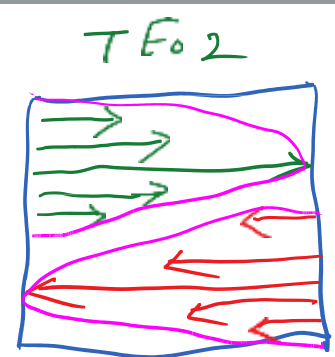
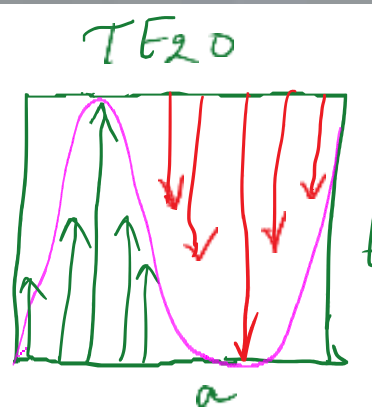
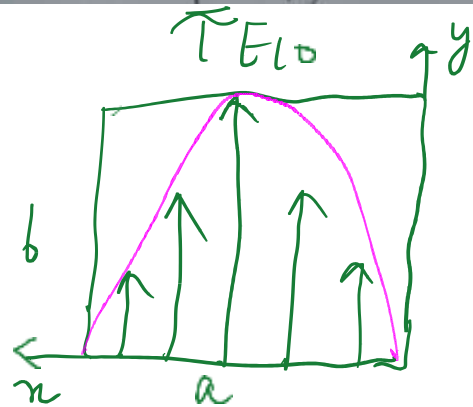
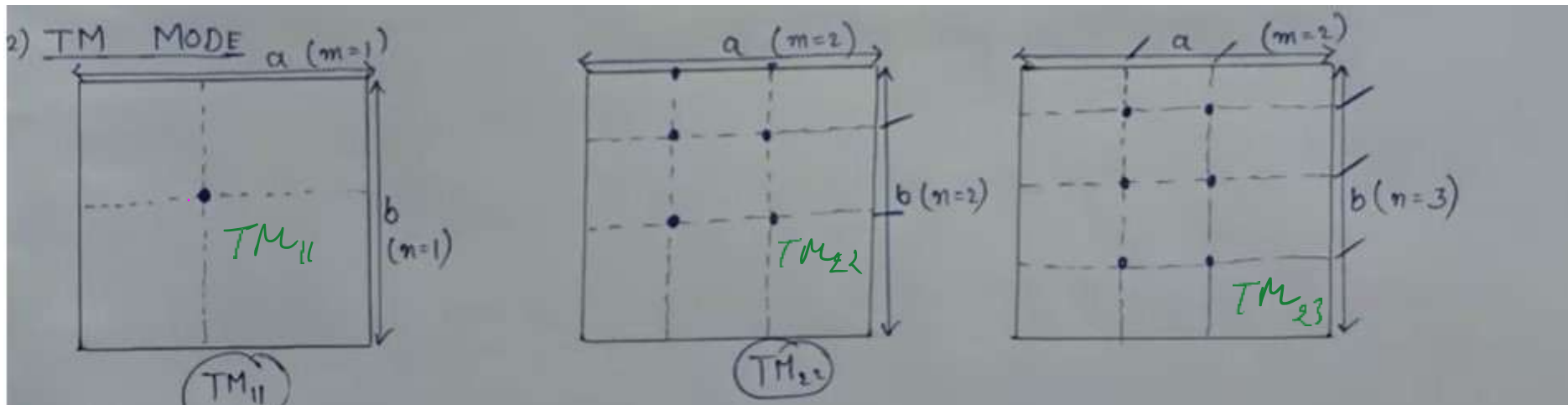
H field in rectangular waveguide TE₁₀



Ey field in rectangular waveguide TE₂₀



Field patterns in waveguide



Rectangular Waveguide (TE) $E_z = 0$



TE_{11}

$$E_x = \frac{j\omega\mu}{h^2} C \left(\frac{n\pi}{b} \right) \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-j\beta'_2 z}$$

$$E_y = -\frac{j\omega\mu}{h^2} C \left(\frac{m\pi}{a} \right) \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-j\beta'_2 z}$$

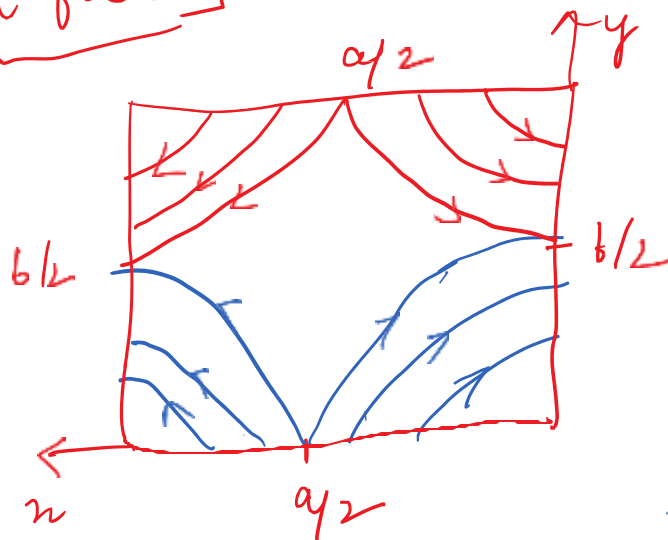
$$H_x = -\frac{j\beta'_1}{h^2} C \left(\frac{m\pi}{a} \right) \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-j\beta'_2 z}$$

$$H_y = -\frac{j\beta'_1}{h^2} C \left(\frac{n\pi}{b} \right) \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-j\beta'_2 z}$$

$$\checkmark H(z) = C e^{-j\beta'_2 z} \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

TE_{11} mode

E field



$n=1$



E_x $m=1$

$n=1$

$$\left[\begin{array}{l} E_x \rightarrow \text{at } x=a/2 \\ E_x=0 \\ \text{at } y=b/2 \quad E_x \text{ is max} \end{array} \right]$$

$$E_x = C_1 \cos\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi y}{b}\right) e^{-j\beta z}$$

$$E_y = C_2 \sin\left(\frac{\pi x}{a}\right) \cos\left(\frac{\pi y}{b}\right) e^{-j\beta z}$$

$E_y \rightarrow$ at $x=a/2$ max
at $y=b/2$ min

A rectangular waveguide has a cross-section of $1.5 \text{ cm} \times 0.8 \text{ cm}$, $\underline{\sigma} = 0$, $\underline{\mu} = \underline{\mu}_0$ & $\epsilon = 4\epsilon_0$. The magnetic field component is given as

$$H_y = 2 \sin\left(\frac{\pi x}{a}\right) \cos\left(\frac{3\pi y}{b}\right) \sin(\pi \times 10^{11} t - \beta z)$$

A/m

Find

- a) The mode of operation.
- b) The cut off frequency
- c) The phase constant (β')

Solⁿ

$$H_m = 2 \sin \left(\frac{\pi n}{a} \right) \cos \left(\frac{3\pi y}{b} \right) \sin(\pi \times 10^{11} t - \beta z)$$

$m=1$ $n=3$

A/m

a) So TE₁₃ or TM₁₃ mode

b) $f_c = \frac{c'}{2} \sqrt{\frac{1}{a^2} + \frac{9}{b^2}}$

$$a = 1.5 \text{ cm}$$

$$b = 0.8 \text{ cm}$$

$$= \frac{1.5 \times 10^8}{2} \sqrt{\frac{1}{(1.5)^2} + \frac{9}{(0.8)^2}} \quad c' = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{2} = 1.5 \times 10^8 \text{ m/s}$$

$$f_c = \underline{28.57 \text{ GHz}}$$

c)

$$\beta' = \sqrt{\omega^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

$$= \sqrt{\omega^2 \mu \epsilon - \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right)}$$

$$\lambda_c = \frac{2}{\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}}$$

$$\omega = \pi \times 10^{11}$$

$$f = 50 \text{ GHz}$$

$$= \omega \sqrt{\mu \epsilon} \sqrt{1 - \frac{\pi^2}{\omega^2 \mu \epsilon} \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right)}$$

$$\beta' = \beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$= \omega \sqrt{\mu \epsilon} \sqrt{1 - \frac{\pi^2 c^2}{4\pi^2 f^2} \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right)}$$

$$\beta' = \omega \sqrt{\mu \epsilon} \sqrt{1 - \frac{\lambda^2}{\lambda_c^2}}$$

$$\text{or } = \omega \sqrt{\mu \epsilon} \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$\beta' = \frac{\pi \times 10^{11} \times 2}{3 \times 10^8} \sqrt{1 - \left(\frac{28.57}{50}\right)^2} = 1718.81 \text{ rad/m}$$

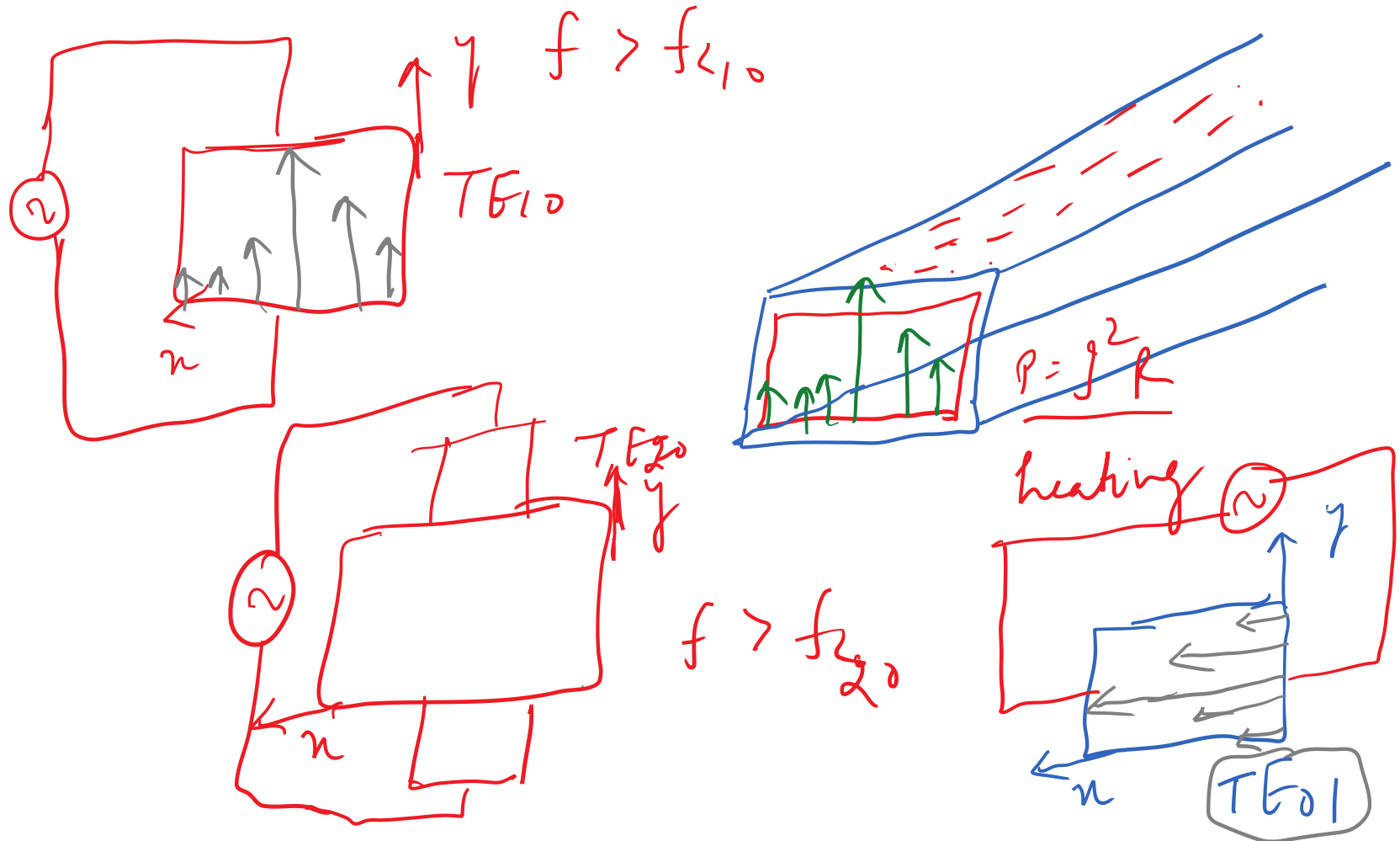
$$\beta' = \beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$v_p' = \frac{\omega}{\beta'} = \frac{\omega}{\beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

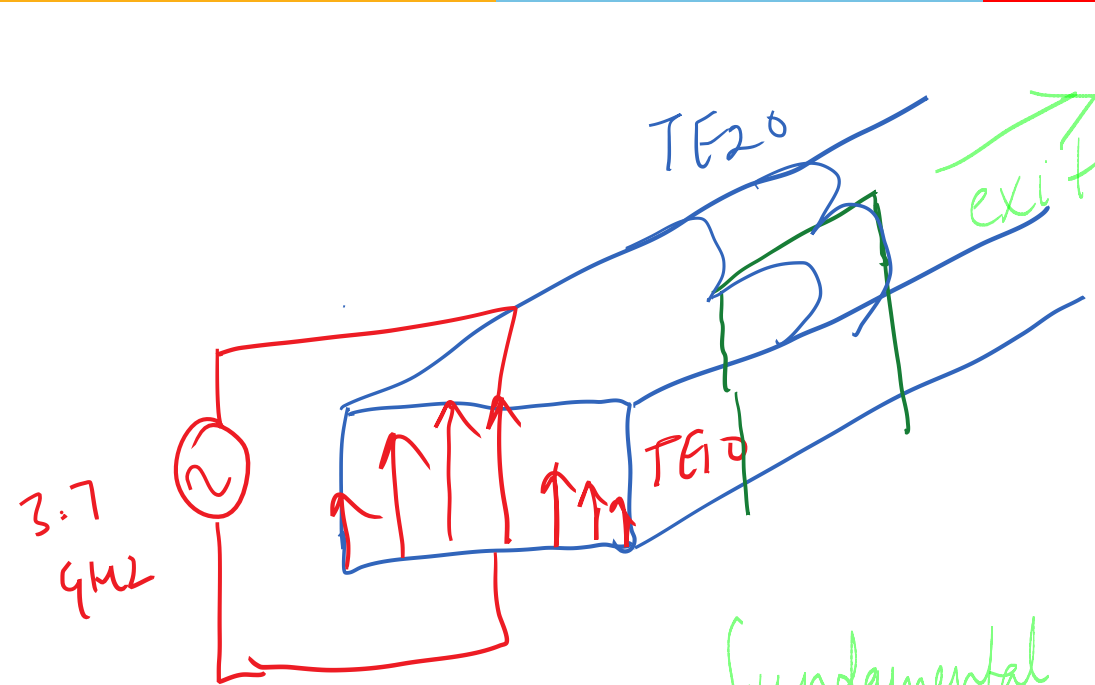
$$v_p' = \frac{\cancel{\omega} \sqrt{\mu \epsilon}}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

$$v_p' = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

Rectangular Waveguide



How do the modes propagate in Rectangular Waveguide



$$f_{c01} = 3.3 \text{ GHz}$$

$$f_{c02} = 4 \text{ GHz}$$

$$f_c > \underline{f_{c20}}$$

then only possible at the exit

$$\underline{\underline{TE_{20}}}$$

fundamental
↑ mode

$$f_{TE_{01}} < f < \underline{f_{TE_{20}}}$$

TEM mode in rectangular waveguide

innovate

achieve

lead

$\underline{TE_{mn}}$

1 0

0 1

1 1

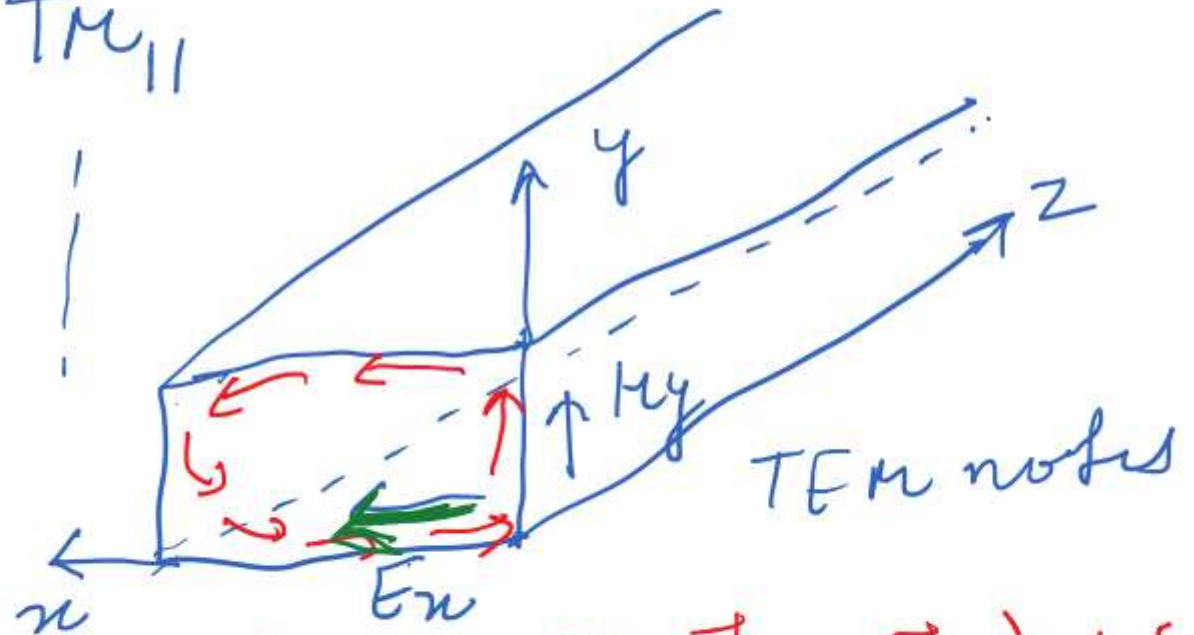
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TM_{mn}

TM_{11}

$TM_0 \rightarrow TEM$



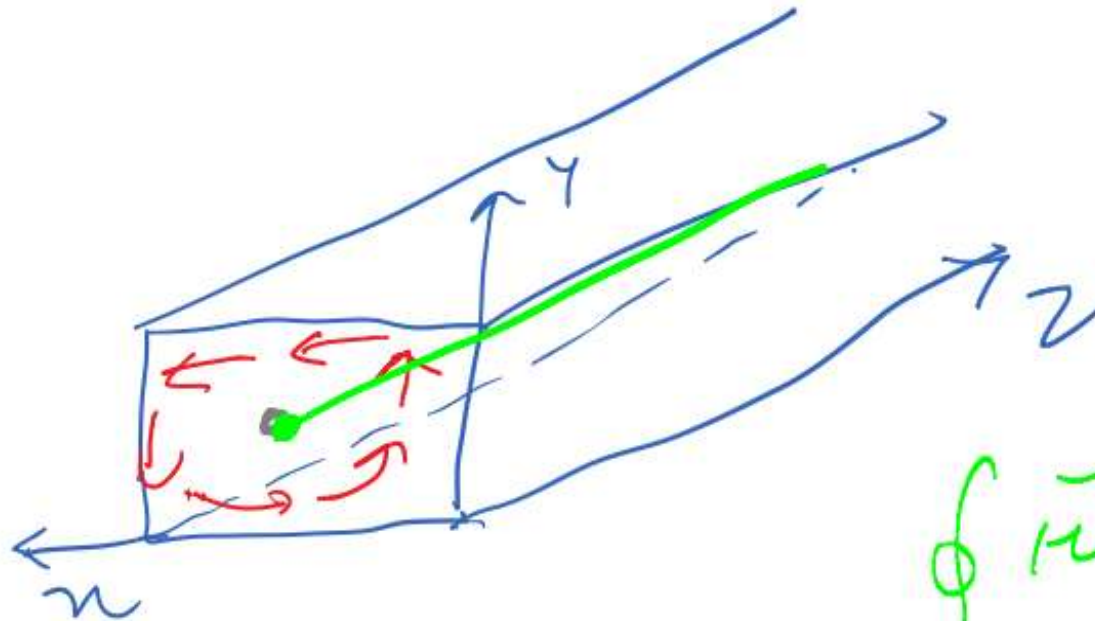
Amperis law = $\oint \vec{H} \cdot d\vec{l} = \int \int (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{S}$

No H exists.

$\oint \vec{H} \cdot d\vec{l} = 0$

No TEM mode.

TEM mode in two conductor system



$$\oint \vec{H} \cdot d\vec{l} = I$$

so Two conductor system $\rightarrow$
coaxial
cable

Modes comparison

innovate

achieve

lead

TEM

① No cut off freq.

② $\beta = \omega \sqrt{\mu\epsilon}$

Non Dispersive

③ No TEM mode in rectangular waveguide.

★★ High power signal or less attenuation waveguide is best option.

TE/TM

① $f < \rightarrow$ cut off freq.

② $\beta' = f(m, n) = \frac{2\pi}{\lambda_g}$

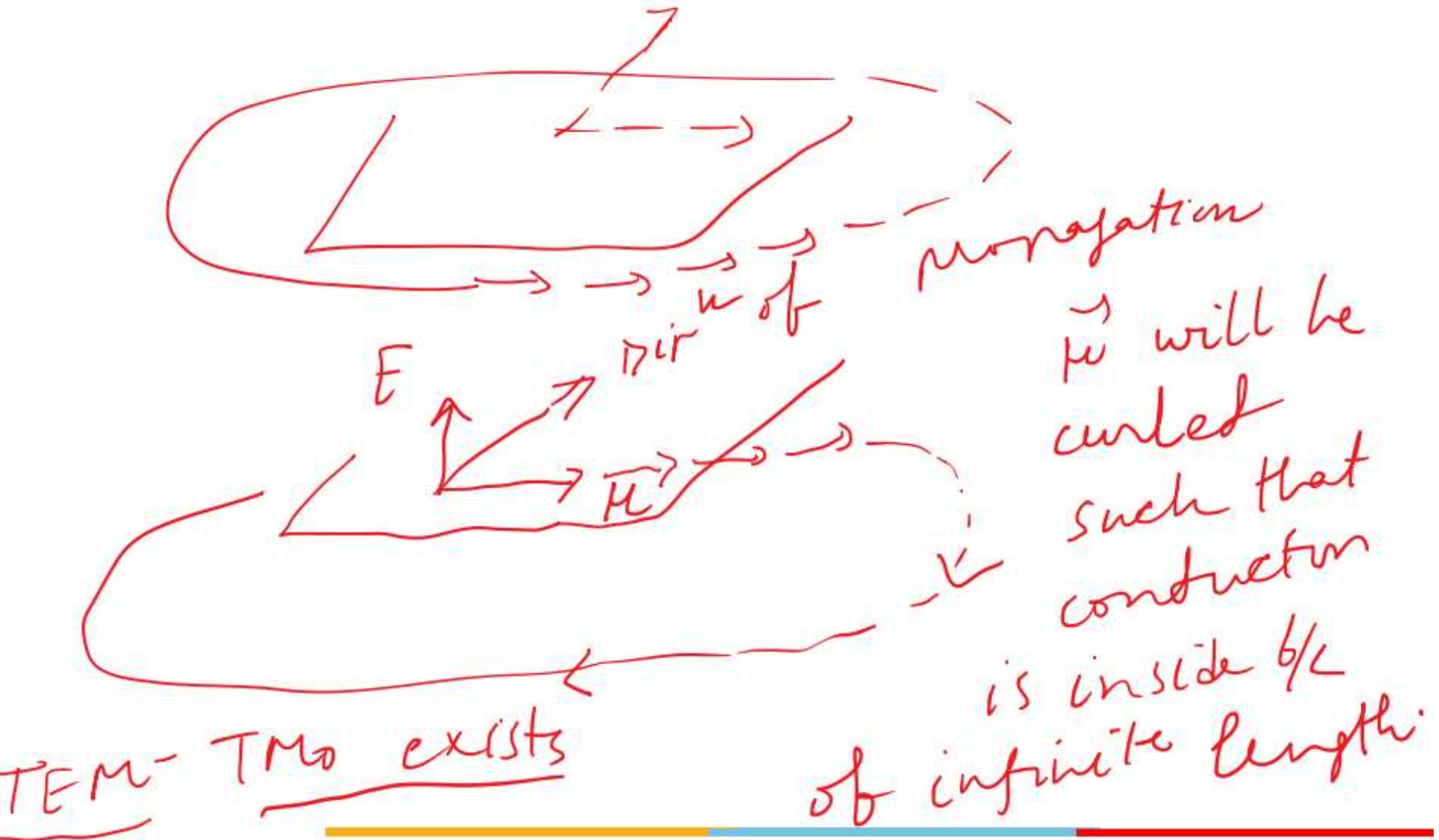
③ Dispersive
TE & TM modes in rectangular waveguide

TEM mode in parallel plate waveguide

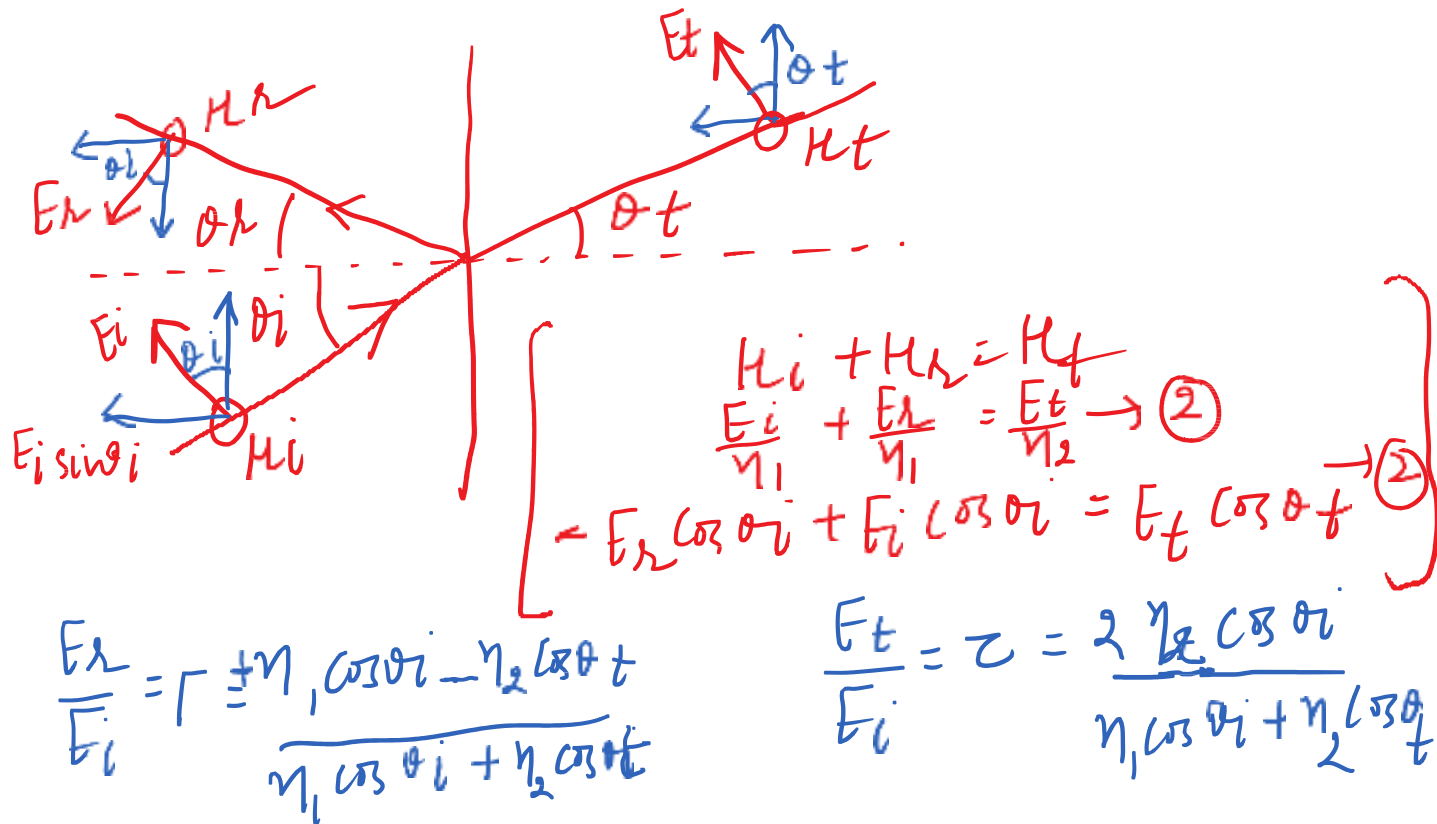
innovate

achieve

lead



Parallel polarization: TM



$$\frac{E_r}{E_i} = \frac{\eta_1 \cos \theta_i - \eta_2 \cos \theta_t}{\eta_1 \cos \theta_i + \eta_2 \cos \theta_t}$$

TM polarization
what if $\underline{\Gamma_{11} = 0}$

$$\underline{n_1 = n_2}$$

$$\eta_1 \cos \theta_i = \eta_2 \cos \theta_t$$

$$\eta_1^2 \cos^2 \theta_i = \eta_2^2 (1 - \sin^2 \theta_t)$$

$$\eta_1^2 \cos^2 \theta_i = \eta_2^2 \left(1 - \frac{\epsilon_1}{\epsilon_2} \sin^2 \theta_i \right)$$

$$\eta_1^2 (1 - \sin^2 \theta_i) = \eta_2^2 - \eta_2^2 \frac{\epsilon_1}{\epsilon_2} \sin^2 \theta_i$$

$$\eta_1^2 - \eta_1^2 \sin^2 \theta_i = \eta_2^2 - \frac{\mu_2}{\epsilon_2} \frac{\epsilon_1}{\epsilon_2} \sin^2 \theta_i$$

$$\frac{\mu_1}{\epsilon_1} - \frac{\mu_1}{\epsilon_1} \sin^2 \theta_i = \frac{\mu_2}{\epsilon_2} - \frac{\mu_2 \epsilon_1}{\epsilon_2^2} \sin^2 \theta_i$$

$$\left[\begin{aligned} n_1 \sin \theta_i &= n_2 \sin \theta_t \\ \sqrt{\mu_1 \epsilon_1} \sin \theta_i &= \sqrt{\mu_2 \epsilon_2} \sin \theta_t \\ \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sin \theta_i &= \sin \theta_t \end{aligned} \right]$$

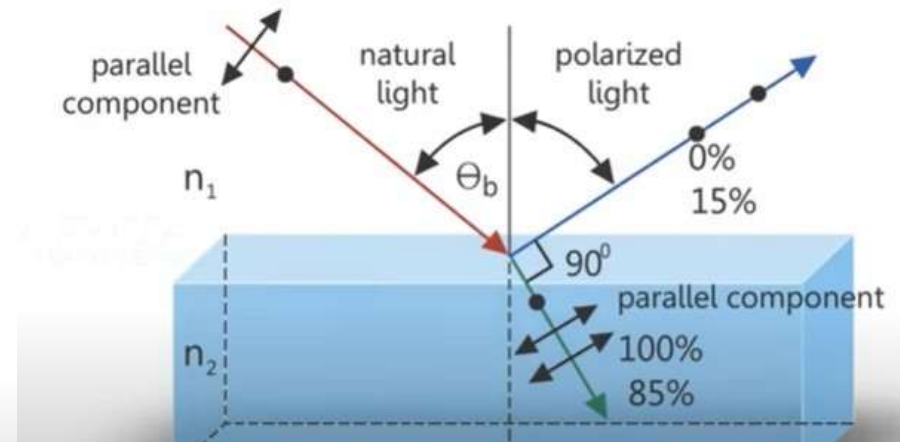
$$\left(\frac{\epsilon_2}{\epsilon_1} - 1 \right) \cos^2 \theta_i = \left(1 - \frac{\epsilon_1}{\epsilon_2} \right) \sin^2 \theta_i$$

$$\tan^2 \theta_i = \frac{\epsilon_2 - \epsilon_1}{\epsilon_1} \times \frac{\epsilon_2}{\epsilon_2 - \epsilon_1}$$

$$\boxed{\tan^2 \theta_i = \frac{\epsilon_2}{\epsilon_1}} \quad \tan \theta_i = \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \frac{n_2}{n_1}$$

Polarization by Reflection

- When an unpolarized light beam is reflected from a surface, the reflected light may be
 - Completely polarized
 - Partially polarized
 - Unpolarized
- It depends on the angle of incidence
 - If the angle is 0° , the reflected beam is unpolarized
 - For other angles, there is some degree of polarization
 - For one particular angle, the beam is completely polarized



Brewster's angle or angle of polarization is defined as the angle of incidence at which light with a specific polarization (TM) is perfectly transmitted without any reflection.

Perpendicular polarization: TE

$$\frac{E_r}{E_i} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} = \Gamma_{\perp}$$

Reflection coefficient
for $\Gamma_{\perp} = 0$

$$\eta_2 \cos \theta_i = \eta_1 \cos \theta_t$$

$$\eta_2^2 (1 - \sin^2 \theta_i) = \eta_1^2 (1 - \sin^2 \theta_t)$$

$$\frac{\mu_2}{\epsilon_2} (1 - \sin^2 \theta_i) = \frac{\mu_1}{\epsilon_1} \left(1 - \frac{\mu_1 \epsilon_1 \sin^2 \theta_i}{\epsilon_2 \mu_2} \right)$$

$$\frac{1}{\epsilon_2} (1 - \sin^2 \theta_i) = \frac{1}{\epsilon_1} \left(1 - \frac{\epsilon_1}{\epsilon_2} \sin^2 \theta_i \right)$$

$$\mu_1 = \mu_2$$

$$\frac{1}{\epsilon_2} - \frac{1}{\epsilon_2} \sin^2 \theta_i = \frac{1}{\epsilon_1} - \frac{1}{\epsilon_2} \sin^2 \theta_i$$

θ_i Does not exist